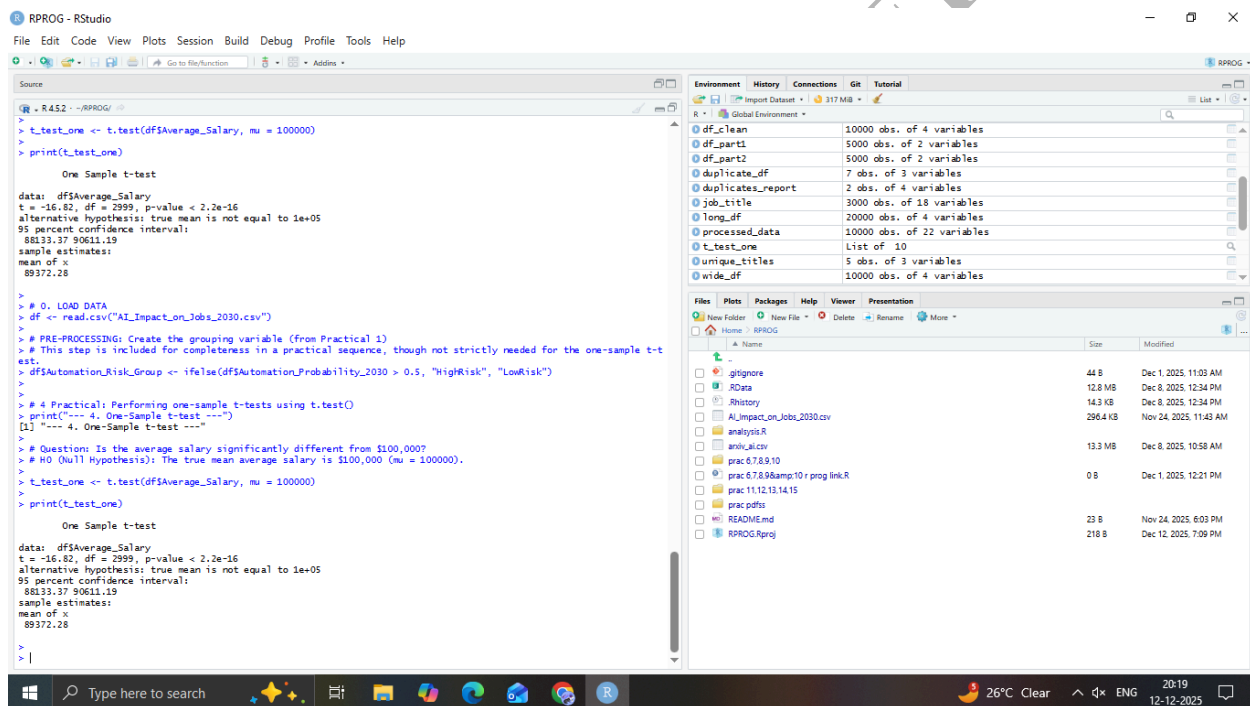


SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME : DATA ANALYSIS WITH SAS/SPSS/R

PRACTICAL 4

AIM: Performing one-sample t-tests using t.test()



The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for a one-sample t-test. The code reads a CSV file, performs a t-test, and prints the results. The output shows a t-value of -16.82, a p-value less than 2.2e-16, and a 95% confidence interval of [88133.37, 90611.19].
- Environment:** Lists the objects in the global environment, including 'df_clean' (10000 obs. of 4 variables), 'df_part1' (5000 obs. of 2 variables), 'df_part2' (5000 obs. of 2 variables), 'duplicate_df' (7 obs. of 3 variables), 'duplicates_report' (2 obs. of 4 variables), 'job_title' (3000 obs. of 18 variables), 'long_df' (20000 obs. of 4 variables), 'processed_data' (10000 obs. of 22 variables), 't_test_one' (List of 10), 'unique_titles' (5 obs. of 3 variables), and 'wide_df' (10000 obs. of 4 variables).
- Files:** Shows the file explorer with various files and folders, including 'Rhistory', 'AI_Impact_on_Jobs_2030.csv', 'analysis.R', 'anish_ai.csv', 'prac 6.7.8.9.10', 'prac 6.7.8.9&10 r prog link.R', 'prac 11.12.13.14.15', 'prac pdfs', 'README.md', and 'RPROG.Rproj'.

```
> t_test_one <- t.test(df$Average_Salary, mu = 100000)
> print(t_test_one)

One Sample t-test

data: df$Average_Salary
t = -16.82, df = 2999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 1e+05
95 percent confidence interval:
 88133.37 90611.19
sample estimates:
mean of x
 89372.28

> # 0. LOAD DATA
> df <- read.csv("AI_Impact_on_Jobs_2030.csv")
>
> # PRE-PROCESSING: Create the grouping variable (from Practical 1)
> # This step is included for completeness in a practical sequence, though not strictly needed for the one-sample t-test.
> df$Automation_Risk_Group <- ifelse(df$Automation_Probability_2030 > 0.5, "HighRisk", "LowRisk")
>
> # 4 Practical: Performing one-sample t-tests using t.test()
> print("---- 4. One-Sample t-test ----")
[1] "---- 4. One-Sample t-test ----"
>
> # Question: Is the average salary significantly different from $100,000?
> # H0 (Null Hypothesis): The true mean average salary is $100,000 (mu = 100000).
> t_test_one <- t.test(df$Average_Salary, mu = 100000)
> print(t_test_one)

One Sample t-test

data: df$Average_Salary
t = -16.82, df = 2999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 1e+05
95 percent confidence interval:
 88133.37 90611.19
sample estimates:
mean of x
 89372.28

> |
```

PURVI ROSHAN KARKERA

S086